

Project Design Phase

Proposed Solution Template

Date	30 July 2026
Team ID	Individual / Solo Project
Project Name	Automated Network Request Management in ServiceNow
Student	Bandhanadam Antony
Maximum Marks	2 Marks

The project team shall fill the following information in the proposed solution template to outline the core problem, the envisioned workflow, and the validation methodology.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Employees currently request network access and firewall changes through email or informal IT channels. This leads to configuration delays, zero visibility into request status, inconsistent InfoSec approvals, and severe security compliance risks due to a lack of an audit trail.
2	Idea / Solution Description	A standardized Service Catalog item in ServiceNow allows employees to formally request network access. The request is automatically routed to managers and InfoSec for approval, checked against active CMDB topologies, and seamlessly fulfilled via IntegrationHub (e.g., Ansible) with the employee notified at each stage.
3	Novelty / Uniqueness	Combines user-friendly catalog ordering with real-time risk assessment, CMDB validation, and zero-touch remote network fulfillment in a single unified workflow. It transitions IT from manual CLI typing to a fully governed, automated orchestration process.
4	Social Impact / Customer Satisfaction	Employees obtain critical network access faster and can independently track status. Reduced back-and-forth communication lowers frustration for developers and Network Admins alike, while automated security validations protect the organization from breaches.
5	Scalability of the Solution	Built upon ServiceNow's multi-instance cloud architecture, the Flow Designer workflow scales to accommodate any volume of network endpoints, security policies, and organizational departments without requiring infrastructure re-architecture.

6	Technical Validation & Debugging
To ensure the integrity of the solution, the following debugging and validation practices are implemented:	

- **Flow Execution Logs:** Used Flow Designer's Execution Details to trace flow steps, identify failures, and verify condition paths during approvals and notifications.
- **Email Logs:** Checked email logs in System Logs → Email → Sent/Received to validate notification triggers and message delivery status.

- **Variable Population Issues:** Resolved misaligned or missing data bindings using Variable set debugging and Catalog client scripts for dynamic population.
- **Form Behaviour Testing:** Confirmed UI policies and visibility logic via Service Portal form tests and preview mode.